

Online Course Enrollment System

pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0" ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.enrollment</groupId>
<artifactId>course-enrollment-system</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>war</packaging>
<properties>
<java.version>17</java.version>
</properties>
<dependencies>
<!-- Spring MVC -->
<dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-webmvc</artifactId>
    <version>6.1.4</version>
</dependency>
<!-- Servlet API -->
<dependency>
<groupId>jakarta.servlet</groupId>
<artifactId>jakarta.servlet-api</artifactId>
<version>6.0.0</version>
<scope>provided</scope>
```

```
</dependency>
<!-- JSP & JSTL -->
<dependency>
<groupId>jakarta.servlet.jsp.jstl</groupId>
<artifactId>jakarta.servlet.jsp.jstl-api</artifactId>
<version>3.0.0</version>
</dependency>
<dependency>
<groupId>org.glassfish.web</groupId>
<artifactId>jakarta.servlet.jsp.jstl</artifactId>
<version>3.0.1</version>
</dependency>
</dependencies>
<build>
<plugins>
<!-- Compiler Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
<configuration>
<source>17</source>
<target>17</target>
</configuration>
```

/plugin>

</plugins>

</build>

</project>

web.xml:

<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee" version="6.0">

<display-name>Online Course Enrollment</display-name>

<servlet>

<servlet-name>dispatcher</servlet-name>

<servlet-

class>org.springframework.web.servlet.DispatcherServlet</servlet-
class>

<init-param>

<param-name>contextConfigLocation</param-name>

<param-value>/WEB-INF/dispatcher-servlet.xml</param-value>

</init-param>

<load-on-startup>1</load-on-startup>

</servlet>

<servlet-mapping>

<servlet-name>dispatcher</servlet-name>

<url-pattern>/</url-pattern>

</servlet-mapping>

</web-app>

Course.java:

```

@Controller
public class CourseController {

    @GetMapping("/courses")
    public String showCourses(Model model) {
        List<Course> courses = List.of(
            new Course(1, "Spring Boot"),
            new Course(2, "Java Fundamentals"));
        model.addAttribute("courses", courses);
        return "courses";
    }

    @GetMapping("/enroll")
    public String showEnrollForm(Model model) {
        model.addAttribute("student", new Student());
        return "enroll";
    }

    @PostMapping("/submitEnrollment")
    public String processEnrollment(@ModelAttribute Student student,
        Model model) {
        model.addAttribute("student", student);
        return "success";
    }
}

```

courseController.java:

```

package com.enrollment.controller;

```

```
import com.enrollment.model.Course;
import com.enrollment.model.Student;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;
import java.util.Arrays;
import java.util.List;

@Controller
public class CourseController {

    // Show list of courses

    @GetMapping("/courses")
    public String showCourses(Model model) {
        List<Course> courses = Arrays.asList(
            new Course(1, "Spring MVC", "Learn Spring MVC fundamentals"),
            new Course(2, "Java 17", "Master Java 17 features"),
            new Course(3, "Hibernate ORM", "Persist data with Hibernate")
        );
        model.addAttribute("courses", courses);
        return "courses"; // JSP: /WEB-INF/views/courses.jsp
    }

    // Show enrollment form

    @GetMapping("/enroll")
    public String showEnrollmentForm(@RequestParam("courseId") int
        courseId, Model model) {
```

```

Student student = new Student();
student.setCourseId(courseId); // Pre-fill selected course
model.addAttribute("student", student);
return "enroll"; // JSP: /WEB-INF/views/enroll.jsp
}

// Handle form submission
@PostMapping("/submitEnrollment")
public String submitEnrollment(@ModelAttribute("student") Student
student, Model model) {
    // Normally, you'd save student info to the database here
    model.addAttribute("student", student);
    return "success"; // JSP: /WEB-INF/views/success.jsp
}
}

```

Student.java:

```

package com.enrollment.model;

public class Student {
    private String name;
    private String email;
    private int courseId;

    // Constructors
    public Student() {
    }

    public Student(String name, String email, int courseId) {
    }
}

```

```
this.name = name;
this.email = email;
this.courseId = courseId;
}
// Getters and Setters
public String getName() {
return name;
}
public void setName(String name) {
this.name = name;
}
public String getEmail() {
return email;
}
public void setEmail(String email) {
this.email = email;
}
public int getCourseId() {
return courseId;
}
public void setCourseId(int courseId) {
this.courseId = courseId;
}
}
```

WebConfig.java:

```
package com.enrollment.config;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.Configuration;
import
org.springframework.web.servlet.config.annotation.EnableWebMvc;
import
org.springframework.web.servlet.config.annotation.ViewResolverReg
istry;
import
org.springframework.web.servlet.config.annotation.WebMvcConfigur
er;
import
org.springframework.web.servlet.view.InternalResourceViewResolver
;

@Configuration
@EnableWebMvc
@ComponentScan(basePackages = "com.enrollment")
public class WebConfig implements WebMvcConfigurer {

    @Override
    public void configureViewResolvers(ViewResolverRegistry registry)
    {

        InternalResourceViewResolver resolver = new
        InternalResourceViewResolver();

        resolver.setPrefix("/WEB-INF/views/"); // location of JSP files
        resolver.setSuffix(".jsp"); // file extension
    }
}
```



```
registry.viewResolver(resolver);  
  
}  
  
}
```

course.jsp:

```
<%@ page contentType="text/html; charset=UTF-8" language="java"  
%>  
  
<%@ page import="java.util.List" %>  
  
<%@ page import="com.enrollment.model.Course" %>  
  
<html>  
  
<head>  
  
<title>Available Courses</title>  
  
</head>  
  
<body>  
  
<h2>Available Courses</h2>  
  
<%  
  
List<Course> courses = (List<Course>)  
request.getAttribute("courses");  
  
if (courses == null || courses.isEmpty()) {  
  
%>  
  
<p>No courses available at the moment.</p>  
  
<%  
  
} else {  
  
for (Course course : courses) {  
  
%>
```

```
<div style="border: 1px solid #aaa; padding: 10px; margin: 10px;">
```

```
<h3><%= course.getName() %></h3>
```

```
<p><strong>Description:</strong> <%= course.getDescription() %></p>
```

```
<form action="enroll" method="get">
```

```
<input type="hidden" name="courseId" value="<%= course.getId() %>">
```

```
<input type="submit" value="Enroll">
```

```
</form>
```

```
</div>
```

```
<%
```

```
}
```

```
}
```

```
%>
```

```
</body>
```

```
</html>
```

enroll.jsp:

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
```

```
<%@ page import="com.enrollment.model.Course" %>
```

```
<html>
```

```
<head>
```

```
<title>Course Enrollment</title>
```

```
</head>
```

```
<body>
```

```

<h2>Enroll in Course</h2>

<%
Course course = (Course) request.getAttribute("course");
if (course == null) {
%>

<p>Invalid course selected.</p>

<%
} else {
%>

<h3>Course: <%= course.getName() %></h3>

<p><strong>Description:</strong> <%= course.getDescription()
%></p>

<form action="submitEnrollment" method="post">

<input type="hidden" name="courseId" value="<%= course.getId()
%>">

<label for="name">Your Name:</label><br>

<input type="text" name="name" required><br><br>

<label for="email">Your Email:</label><br>

<input type="email" name="email" required><br><br>

<input type="submit" value="Submit Enrollment">

</form>

<%
}
%></body>

```

</html>

success.jsp:

```
<%@ page contentType="text/html; charset=UTF-8" language="java"
%>
```

```
<%@ page import="com.enrollment.model.Student" %>
```

```
<%@ page import="com.enrollment.model.Course" %>
```

```
<html>
```

```
<head>
```

```
<title>Enrollment Successful</title>
```

```
</head>
```

```
<body>
```

```
<h2>Enrollment Confirmation</h2>
```

```
<%
```

```
Student student = (Student) request.getAttribute("student");
```

```
Course course = (Course) request.getAttribute("course");
```

```
if (student != null && course != null) {
```

```
%>
```

```
<p>Thank you, <strong><%= student.getName() %></strong>, for
enrolling in <strong><%= course.getName() %></strong>.</p>
```

```
<p>We've sent a confirmation to your email: <strong><%=
student.getEmail() %></strong>.</p>
```

```
<%
```

```
} else {
```

```
%>
```

```
<p>Enrollment information is missing or incomplete.</p><%
```

}

%>

</body>

</html>